

Copenhagen Atomics

thorium breeder reactor

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The goal

Mass manufacturing thorium reactors



Reactor schematics

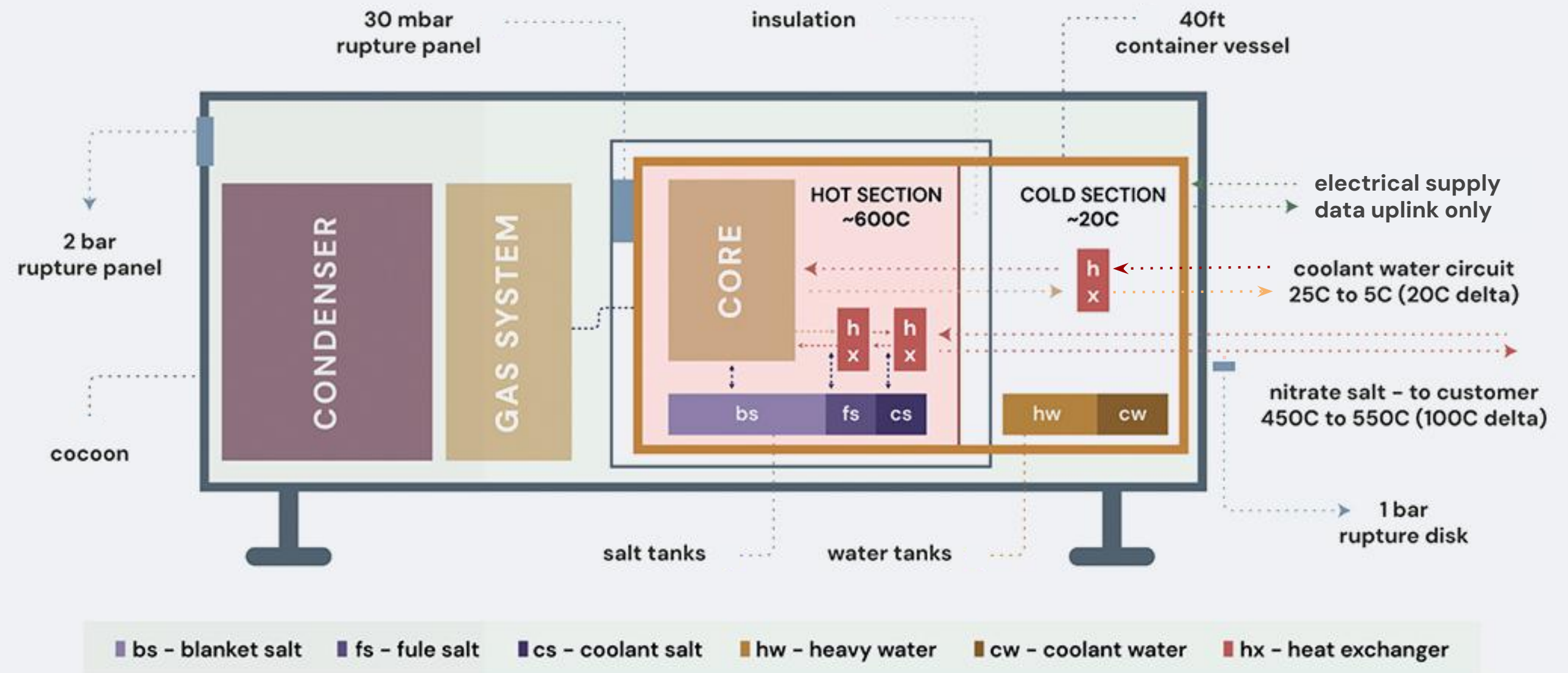
3 barriers of protection, not to scale

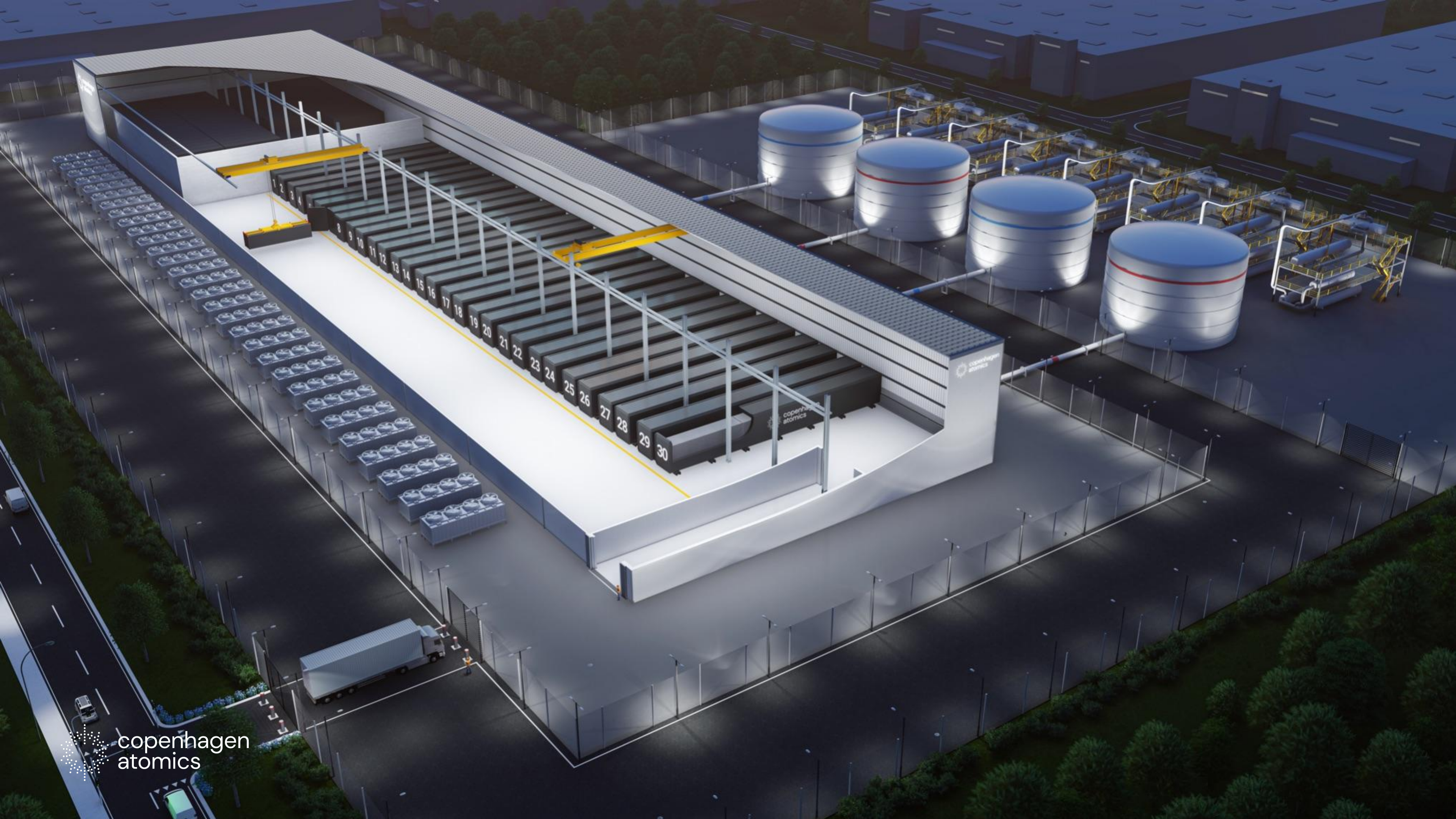
Cocoon provides containment and reactor shielding.

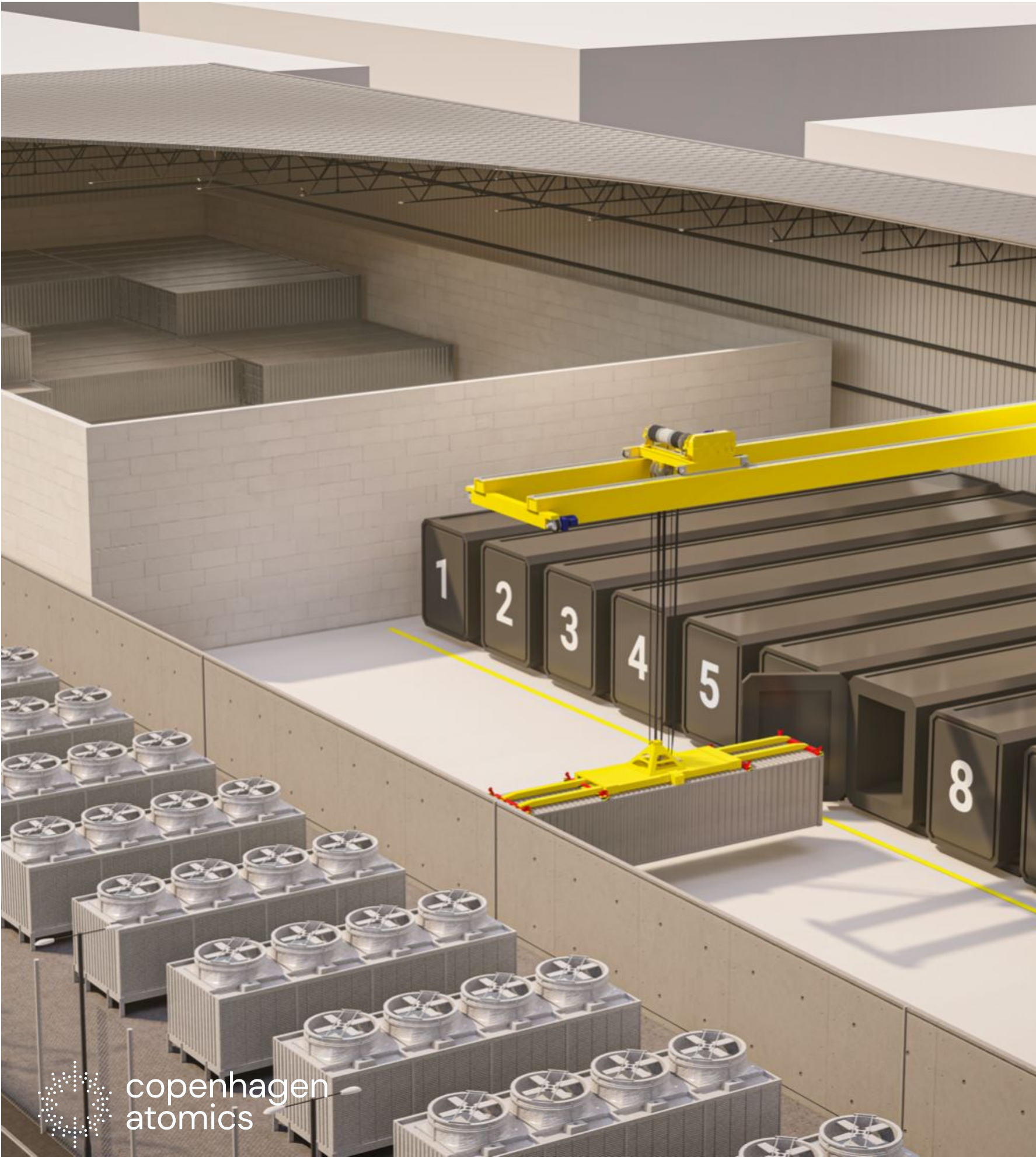
Each 40 foot container vessel is 100% hermetically sealed from production line and until decommissioning.

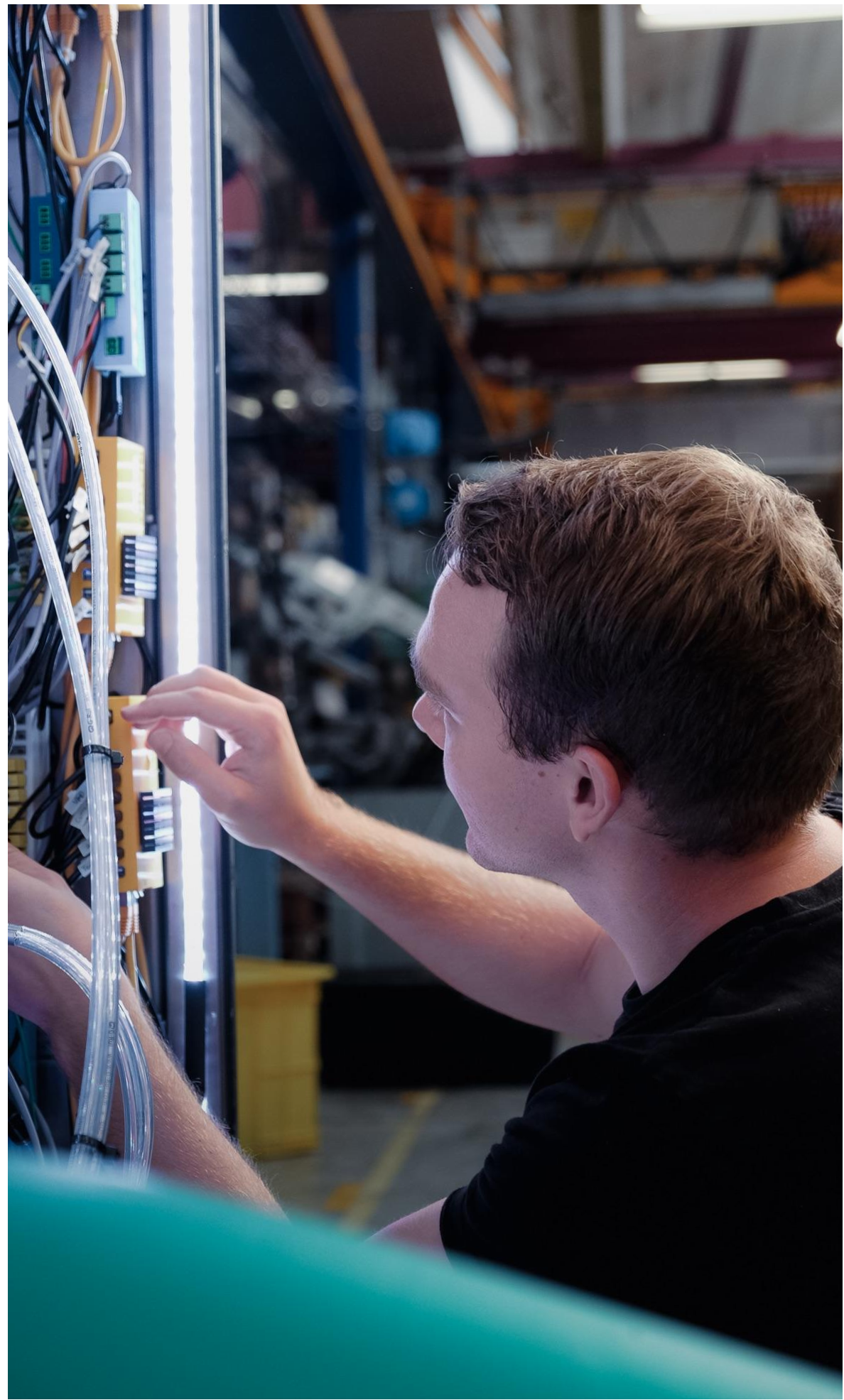
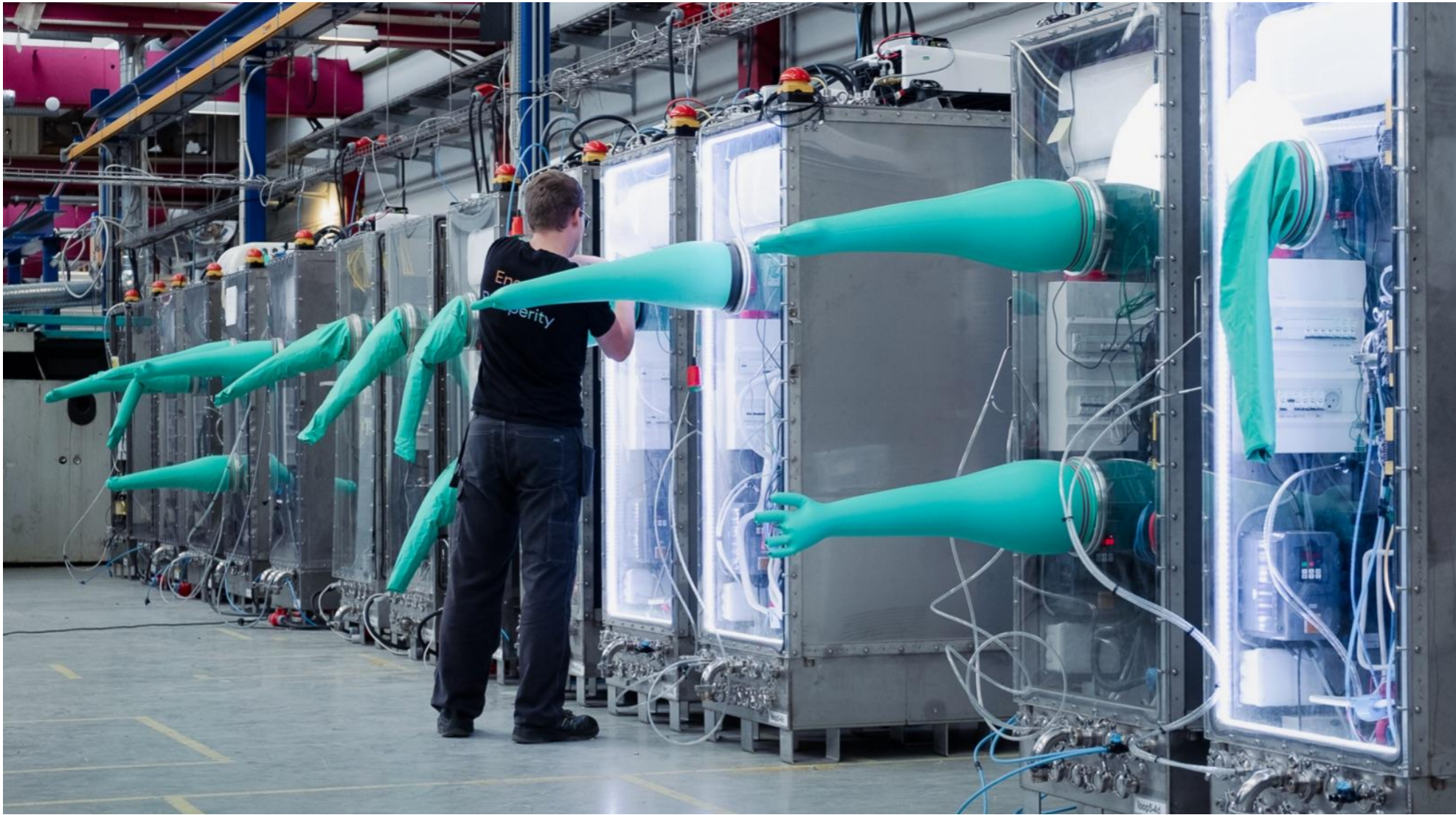
No maintenance, only replacement of reactor vessels.

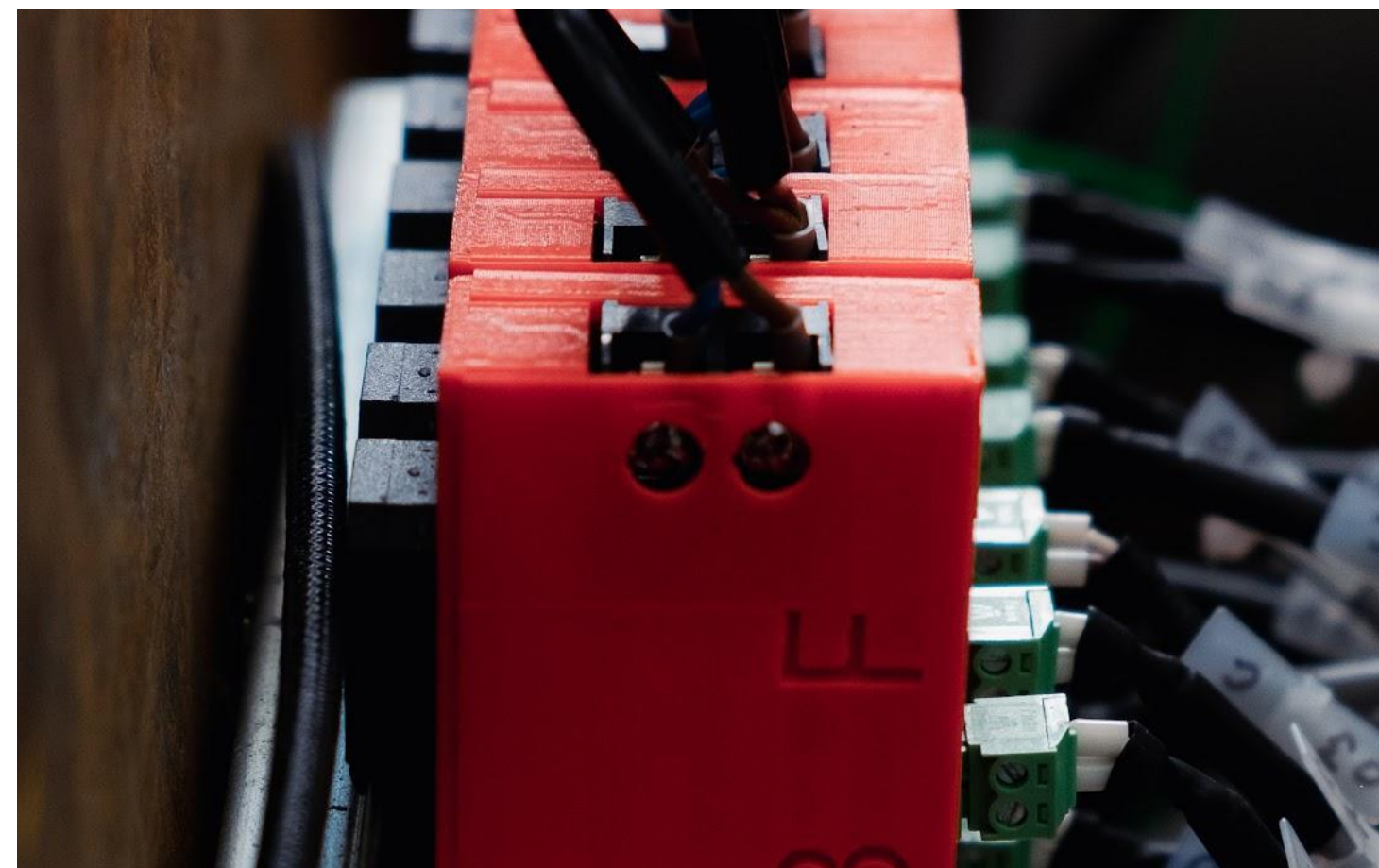
All fuel transfer and fission product separation and storage are confined to within the cocoon, including fission product gasses.









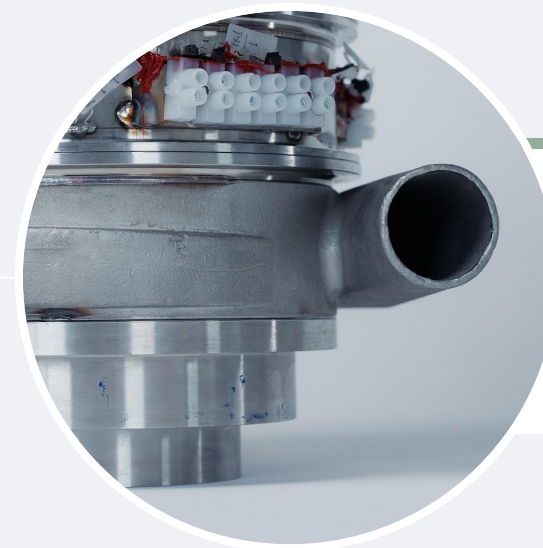




Canned active magnetic bearings keeps the rotor levitation during operation, removing wear and tear.



Canned motor spins the rotor assembly while submerged in molten salt for years without need of maintenance or leakage.



Centrifugal pump which is passively open allowing the salt to quickly drain backwards into storage tanks when the motor shuts down.



